Addendum: R^2 scans over Chris's parameters, $R = 14.3$ Gpc

26 September 2026

1 Variables

Symbol	Cell	Meaning	Value used here
R	B29	radius of the source circle about Earth (limit of vision)	14.3 Gpc $= 4.4127 \times 10^{23}$ km
D/R	C23	CoE distance / shell radius	scanned (sheet: 5)
D	B28	Earth-to-CoE distance, = B29 · C23	$(D/R)\,R$
L/D	D23	density scale length / CoE distance	scanned (sheet: 0.18)
L	B30	scale length of $\rho \propto e^{-r/L}$, = B28 · D23	$(L/D)\,D$
M	B27	mass inside D	scanned (sheet: 4.44×10^{68} kg)
k	B25	expansion rate	2.18×10^{-18} s^{-1} (fixed)
G, dt	B26, B33	gravity constant, time step	6.67×10^{-17}, 1 (fixed)
Amp, ϕ	BA57, BA59	amplitude and phase of the sine factor $1 - \mathrm{Amp}\sin(\theta + \phi)$	scanned (sheet: -0.06, 0.35 rad)
Γ	–	gravity strength $G\,M\,dt/(D^2 k R)$	derived
ℓ_0	–	optional axis offset, $\theta = \ell - \ell_0$; not in the sheet	optional

2 Method

The model is column BH of the tab *Copy of Copy of Copy of density*. The sheet's formulas are kept exactly, including $\mathsf{T} = \mathsf{R} + \mathsf{K}\,dt$. The only change is B29 $= R$; D and L follow from C23 and D23 through B28 and B30, as in the sheet. The fit quality is

$$R^2 = 1 - \frac{\sum_i (y_i - \mathrm{BH}_i)^2}{\sum_i (y_i - \bar{y})^2}, \tag{1}$$

with no refit. The targets are the 71 Migkas points at $\theta = \ell$, $A_F(\ell)$ on a 1° grid, and (secondary) the sheet's lookup-mapped Migkas column AG. The optimiser is a random screen followed by 25 Nelder–Mead starts. Bounds are $1.05 \le D/R \le 10$, $0.02 \le L/D \le 3$, $0.1 \le M/M_{\mathrm{sheet}} \le 1000$ (extended from 10×), $|\mathrm{Amp}| \le 0.15$, and $\phi \in [0, 2\pi)$.

BH depends only on D/R, L/D, Amp, ϕ and Γ. Since $\Gamma = GMdt/[(D/R)^2 kR^3]$, changing R at fixed D/R and L/D is exactly equivalent to rescaling M by $(R_{\mathrm{new}}/R_{\mathrm{old}})^3 = (4.4127/1.3)^3 = 39.11$. k acts like $1/M$ and is not scanned.

3 Results

Case	C23	D23	D (Gpc)	L (Gpc)	M (kg)	M/M_{crit}	Amp	ϕ	ℓ_0	Γ	R^2_{Mig}	$R^2_{A_F}$	R^2_{map}
Sheet values	5	0.18	71.5	12.9	4.44×10^{68}	1.1×10^{12}	−0.06	0.35	–	0.0063	0.266	0.295	0.082
Best vs Migkas	2.26	0.295	32.3	9.51	2.48×10^{69}	6.5×10^{13}	−0.047	0.21	–	0.173	**0.759**	0.833	0.579
Best vs A_F	2.10	0.291	30.0	8.71	1.99×10^{69}	6.5×10^{13}	−0.046	0.16	–	0.161	0.759	**0.834**	0.573
Best vs sheet map	3.59	0.069	51.4	3.56	5.80×10^{69}	3.8×10^{13}	−0.060	0.83	–	0.160	0.588	0.641	**0.790**
Optional, with axis offset ℓ_0:													
Best vs Migkas	2.45	0.297	35.0	10.4	3.54×10^{69}	7.3×10^{13}	−0.042	0.45	13.0°	0.210	**0.897**	0.981	–
Best vs A_F	2.32	0.295	33.1	9.77	3.01×10^{69}	7.3×10^{13}	−0.041	0.41	12.9°	0.199	0.896	**0.982**	–

Table 1: Best fits at $R = 14.3$ Gpc, with ϕ in rad. In km, D is 2.206×10^{24} (sheet), 9.96×10^{23} (best vs Migkas) and 1.081×10^{24} (with ℓ_0); L is 3.97×10^{23}, 2.93×10^{23} and 3.21×10^{23}. $M_{\mathrm{crit}} = \frac{4}{3}\pi\rho_c D^3$ uses $H_0 = 70$ km s^{-1}Mpc^{-1} and SI G.

Changing R leaves every optimum unchanged apart from M. For each fitted case, $M_{\mathrm{new}}/M_{\mathrm{old}} = 39.11$ to five digits; D/R, L/D, Amp, ϕ, ℓ_0 and all R^2 values are identical to the $R = 4.21$ Gpc scan. At the sheet's M, however, Γ drops from 0.247 to 0.0063, and R^2 against the points falls from 0.632 to 0.266.

Cell	Sheet	Best	Range with $R^2 \geq R^2_{\max} - 0.01$	Others at sheet values: best value, R^2	
C23 D/R	5	2.26	1.97–3.12	1.51	0.641
D23 L/D	0.18	0.295	0.26–0.30	0.02 (bound)	0.270
B27 M (kg)	4.44×10^{68}	2.5×10^{69}	1.3×10^{69}–5.9×10^{69}	1.5×10^{70}	0.637
BA57 Amp	-0.06	-0.047	-0.053 – -0.038	-0.053	0.275
BA59 ϕ (rad)	0.35	0.21	-0.1 – 0.47	0.63	0.283
ℓ_0 (optional)	0	13°	10–15°	350°	0.276

Table 2: One-parameter profiles against the Migkas points ($\theta = \ell$). The re-optimised profile curves are identical to the earlier scan with M scaled by 39.11. The fixed-others curves change because Γ is 39 times smaller at the sheet's M.

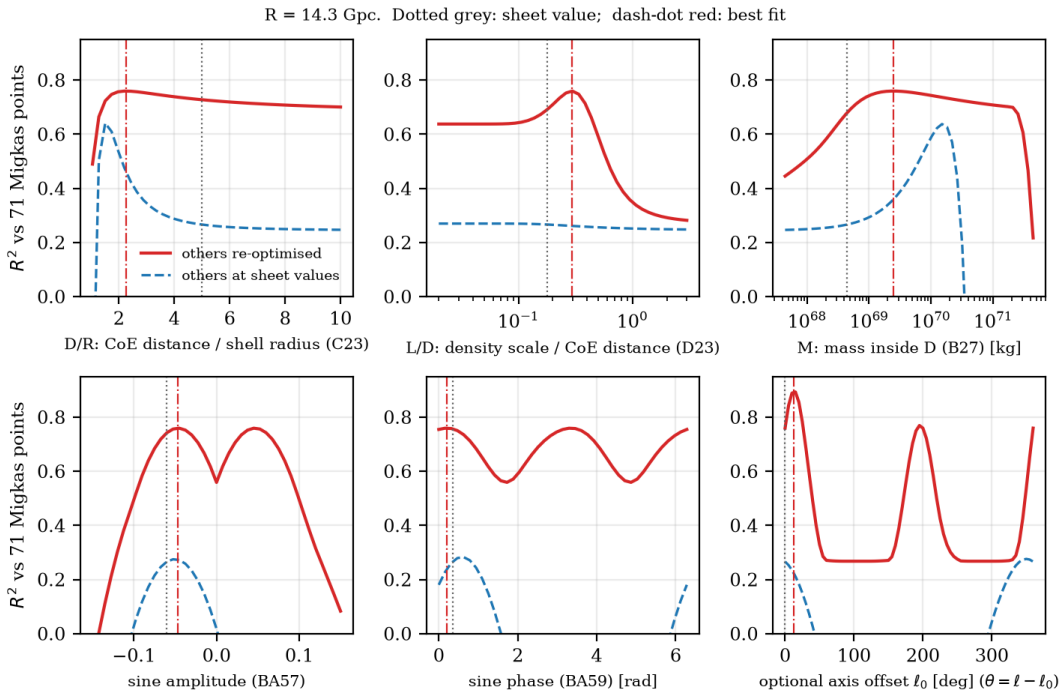


Figure 1: R^2 profiles. Solid: others re-optimised. Dashed: others at sheet values.

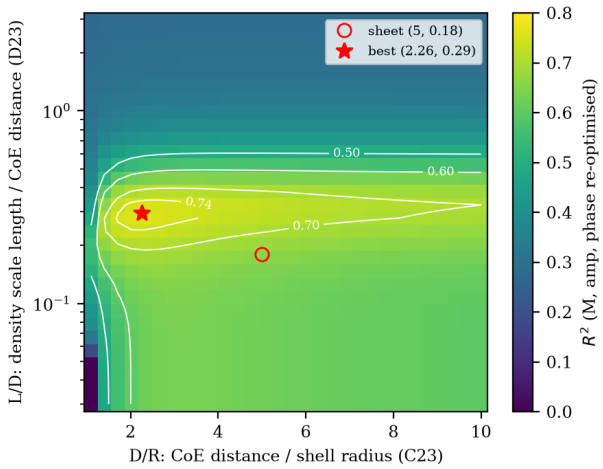


Figure 2: R^2 over D/R and L/D, with M, Amp and ϕ re-optimised ($\theta = \ell$).

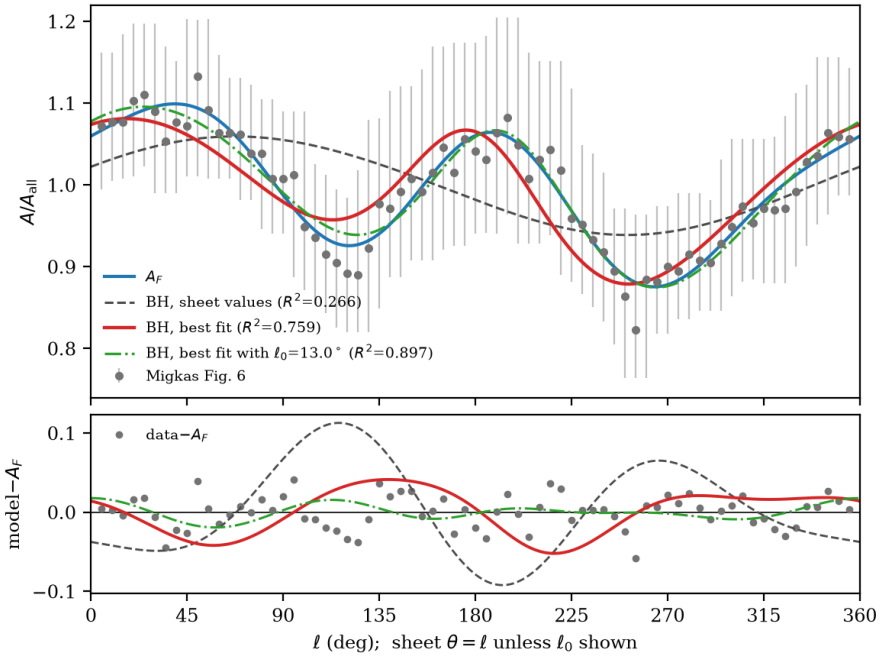


Figure 3: BH at the sheet values ($R = 14.3$ Gpc), at the best fit, and at the best fit with ℓ_0, compared with A_F and the Migkas points.

4 Notes

1. With the bound at $1000\,M_{\text{sheet}}$, no best fit or profile point is limited by M. The largest M reached on any profile is 2.1×10^{71} kg; with the old $10\times$ bound, the D/R profile would have hit it.
2. The D/R–M degeneracy remains: along the D/R profile the best M grows roughly as $(D/R)^3$, and R^2 falls only to 0.70 at $D/R = 10$. $L/D = 0.26$–0.30 is the best-determined parameter.
3. The fitted M is $\sim 10^{13}$–$10^{14}\,M_{\text{crit}}$. This follows from $dt = 1$ s in the sheet: $\Gamma \approx 0.17$ needs $GM/D^2 \approx 0.17\,kR$ per second. If dt were $1/k$, the same Γ would correspond to $M/M_{\text{crit}} = 2\Gamma(k/H_0)^2/(D/R) \approx 0.14$ for the best fit. This is an interpretation, not the sheet. His G is also 10^3 times the SI value in km units.